

2024 CpSc Q6

Section: Software Design and Development

Topic: Implementing a Prime Check — Filling Lines 7 and 43

Question Summary

(a) Write the code for line 7: the condition that tests if the remainder of $n \div \text{divisor}$ equals 0.

(b) Write the code for line 43: set `isPrime` by calling `checkPrime` with the user's input.

Worked Solution

(a) Line 7 (Python)

```
if n % divisor == 0:
```

Explanation: `%` computes the remainder. If the remainder is 0, divisor divides `n` exactly.

(b) Line 43 (Python)

```
isPrime = checkPrime(inputNum)
```

Explanation: we call the function with the keyboard input value and store the returned Boolean.

Common Alternatives

- Java: `if (n % divisor == 0) { ... }` and `boolean isPrime = checkPrime(inputNum);`
- JavaScript: `if (n % divisor === 0) { ... }` and `const isPrime = checkPrime(inputNum);`
- Pseudocode: IF `n MOD divisor = 0` THEN and SET `isPrime` $\leftarrow$ `checkPrime(inputNum)`

Context (Python, for completeness)

```
def checkPrime(n: int) -> bool:
```

```
if n < 2:
    return False
for divisor in range(2, n):
    if n % divisor == 0:
        return False
return True
```

```
inputNum = int(input())
isPrime = checkPrime(inputNum)
print(f"{inputNum} is prime." if isPrime else f"{inputNum}
is not prime.")
```

Final Answer

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- (a) if n % divisor == 0:
- (b) isPrime = checkPrime(inputNum)

Revision Tips

- 'Remainder equals 0' is the universal divisibility test: use MOD (or %) operator.
- Call the function with the variable that holds user input; store the Boolean it returns.
- Edge case: numbers < 2 are not prime (handled before the loop).
- Efficiency tip: you can check up to $\sqrt{n}$, but this is not required for this question.

Exam Alignment

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Matches 2024 MI: line 7 uses modulus comparison with 0;
line 43 calls checkPrime with inputNum and stores the
Boolean.